FECON 490 / ECON 690: Business Economics

Macro Modeling and Industry Implications

Department of Economics, Duke University | Spring 2027

Instructor Information

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Office Hours: See Canvas or by appointment.

Class Schedule and Delivery

Synchronous, in person ‖ Social Sciences 124
6Jan2027–21Apr2027 ‖ Mondays and Wednesdays ‖ 8:30 AM–9:45 AM
The course calendar on Canvas governs meeting-specific topics, activities, and deadlines. There is no class on January 18 or during the March 8–10 break shown in that calendar.

Pre-Requisites

Econometrics: Basic proficiency with cross-sectional linear regression is expected. Familiarity with panel regression is helpful. ECON 204D is the undergraduate prerequisite.

Computing: Basic familiarity with R or Python is recommended. Students may use either language for project research, modeling, and final reproducible code.

Economics and business: Prior familiarity with macroeconomics and basic financial accounting is helpful; relevant concepts will be reviewed as needed.

Course Description

How do macroeconomic conditions shape business revenues, costs, and profits? Which industries are most sensitive to a slowdown or inflation shock? How should decision-makers revise expectations when economic data, monetary policy, or geopolitical conditions change? Business economists address these questions by combining economic theory, data, econometric models, computation, and informed judgment.

Our organizing framework is hierarchical and top down. We begin with five macroeconomic concepts—output, labor markets, inflation, interest rates, and the dollar—and trace their implications through national income, industry profitability, and ultimately individual companies. The central task is to make defensible, internally consistent, next-twelve-month (NTM) projections, together with a risk assessment spanning alternative economic scenarios.

The course has three closely connected units:

1. **Theoretical foundations.** We develop a compact macroeconomic framework; connect macro variables to a national income statement; trace industry and company effects; and introduce scenario analysis.
2. **National activity and empirical calibration.** We work with economic and business data, build and evaluate econometric models for revenues, costs, and profitability, and separate systematic model-based predictions from reasoned adjustments.
3. **Industry and company implications.** We extend the analysis across industries and selected companies, examine heterogeneity in responses to aggregate conditions, and evaluate competing empirical specifications, including machine-learning approaches when useful.

Throughout the course, teams conduct a cumulative consulting exercise called the *Top Down Profit Report* (TDPR). Each team serves as a business-economics consultancy advising clients who understand business decisions but may not have specialized training in macroeconomics or data science. Teams develop evidence, refine models, present and defend conclusions, and revise their outlook as new information arrives.

Analytical and Computational Expectations

The course puts material emphasis on **econometric estimation, empirical validation, and interpretation**. Teams are expected to estimate meaningful relationships that connect the top-down framework to national income, industry performance, or company outcomes; assess model fit, stability, uncertainty, and forecast performance as appropriate; and explain what the evidence does and does not establish. The macro block may be informed by third-party forecasts and economic judgment rather than a fully estimated simultaneous-equation system. Machine learning is encouraged when it improves the research question or its implementation; no particular algorithm or minimum number of ML methods is required. Economic coherence and methodological judgment matter more than model complexity.

Communication and Course Materials

Canvas is the primary source for announcements, readings, assignments, due dates, surveys, grades, and the detailed course calendar. Please check it regularly. There is no required textbook. Lecture materials, including *Profit from the Top Down: A Framework for Connecting Macro to Business Profitability*, will be provided. Optional background readings include:

- *Macroeconomics for Business*, Davidson, Hauskrecht, and von Hagen;
- *Managerial Economics and Business Strategy*, Baye and Prince;
- *A Practical Guide to Macroeconomics*, Jeremy Rudd;
- *The Secrets of Economic Indicators*, Bernard Baumohl;
- *Macroeconomics for Professionals*, Leslie Lipschitz.

Technology

- **Laptops:** Bring a laptop capable of running R or Python and working with course data. We will use statistical computing tools during demonstrations and exercises. Instructor examples may use R; equivalent Python workflows are acceptable for the project.
- **Phones and other devices:** Do not use phones during class unless authorized. Use laptops and tablets for course-related work rather than unrelated activities.
- **Canvas:** Materials, submission instructions, surveys, and due dates will be distributed through the course site. The university's technology support resources can assist with account and platform issues.

Grading

The course centers on a semester-long TDPR exercise. Three teams (A, B, and C) will give brief, cumulative updates; each will also give one substantial in-class presentation. The final submission documents the complete analysis, including reproducible code and data. **Interim presentations are assessed through the evidence and reasoning communicated and defended, not by grading the underlying code. Code is directly graded only with the final submission.**

Undergraduate students (FECON 490)

Assessment	Points (out of 100)
One-slide presentations: February 17, February 24, March 3 (2 each)	6
Full-deck presentation: one date assigned to the team	54
Final comprehensive TDPR, including code and data	30
Class participation	10
Total	**100**

Graduate students (ECON 690)

Assessment	Points (out of 100)
One-slide presentations: February 17, February 24, March 3 (2 each)	6
Full-deck presentation: one date assigned to the team	44
Final comprehensive TDPR, including code and data	25
Class participation	10
Individual Graduate Research Extension	15
Total	**100**

One-slide presentations (6 points total). All teams present one slide on each of February 17, February 24, and March 3. Each update is approximately 5–10 minutes per team, including discussion. These are small, low-stakes opportunities to sharpen the forecast and receive feedback. The April 7 one-slide activity is **ungraded** and does not add points.

Full-deck presentation (54 undergraduate / 44 graduate points). Each team leads one 45–60-minute in-class session, including questions and discussion. Team A presents March 17 (macro, national, one industry); Team B presents March 24 (macro, national, two industries); Team C presents March 31 (macro, national, two industries). Expectations will reflect the scope and material available by the team's assigned date; teams are not penalized merely for presenting earlier. Evaluation focuses on analytical rigor, empirical support, internally coherent forecasts and scenarios, clarity, and the ability to defend decisions. The accompanying slides support the presentation but are not a separately scored written report.

Final comprehensive TDPR (30 undergraduate / 25 graduate points). Each team submits an approximately 60-minute recorded presentation, PDF slides, and the code, data, and documentation needed to reproduce material analysis. The final covers five macro KPIs, the national income statement, two new industries from different sectors, one company per industry, and at least three probability-weighted scenarios. Code and reproducibility are explicitly assessed as part of the final rubric; they are not graded in the interim rounds. Detailed requirements and the final rubric appear in the TDPR exercise document.

Class participation (10 points). This covers preparation, attendance and professionalism, thoughtful questions and discussion, weekly Macro Journals, audience surveys, and engagement with hands-on activities. Journals and surveys are assessed for completion and genuine, reasoned effort rather than the accuracy of the forecast or whether a student agrees with the presenting team. In-class exercises are ungraded practice, although active engagement contributes to participation.

Team and individual scores

The full-deck presentation and final TDPR receive team-level evaluations. Individual scores for each of those major team deliverables combine **25% team score, 25% numeric teammate peer review, and 50% instructor evaluation**. Peer reviews take place after the team's full-deck presentation and again after the final submission. Each student distributes a fixed point budget among teammates, excluding themselves, so that equal contributions receive equal peer scores. Instructor evaluation considers the individual's command of the material, presentation and Q&A contributions, research contributions, and attribution in team materials. Graduate Research Extensions are graded individually. The TDPR exercise document describes the point-budget and normalization details.

The instructor evaluates the quality of analysis, reasoning, implementation when applicable, and communication, rather than whether a forecast ultimately happens to be realized. Final letter-grade thresholds will be determined by the instructor, taking into account the difficulty of the work and overall class performance. Requests for re-evaluation may lead to a higher or lower score upon review of relevant materials.

Graduate Research Extension (ECON 690 Only)

Each graduate student completes an individual, instructor-approved research extension connected to the team's TDPR. The goal is deeper analytical work and independent methodological judgment, not simply more slides or another required machine-learning method. Suitable directions include specification comparisons, forecast validation, structural stability, uncertainty quantification, robustness to data choices, and other substantive questions arising from the team's research. A brief proposal requires instructor approval; timing

and instructions will appear on Canvas. The completed research, code, and conclusions may be integrated into the team's final materials but must be clearly attributable to the graduate student. The extension counts for 15 course points and is graded individually. See the TDPR exercise document for full criteria.

Artificial Intelligence Use

Students may use **Duke-provided ChatGPT Edu and Microsoft Copilot** for appropriate class work, including coding, debugging, understanding concepts, brainstorming, preparing presentations, and individual Macro Journals. Other generative-AI services require prior instructor approval. Students remain responsible for every empirical decision, interpretation, forecast, and submission; AI outputs must be understood, tested, and verified before use. Each team must maintain an **AI Usage Log** of substantive AI interactions throughout the semester. The log is reviewed as part of the final submission, not separately graded at interim presentations. Do not disclose confidential information, passwords, or access tokens to AI services. The TDPR exercise document gives the complete policy and logging instructions.

Tentative Course Schedule

The detailed spreadsheet calendar, as distributed on Canvas, is authoritative for specific meeting topics and dates. The general sequence is:

- **January 6–25:** Introduction, top-down framework, macro modeling, national-income accounting, industry connections, implementation, and risk analysis.
- **January 27–February 8:** Economic data sources and calibration; statistical and computational implementation of the framework.
- **February 10–March 15:** Team research and meetings; all-team one-slide updates on February 17, February 24, and March 3. No class March 8 or 10.
- **March 17–31:** Team A full deck (March 17), Team B full deck (March 24), Team C full deck (March 31), interspersed with team research meetings.
- **April 5–21:** April 7 ungraded one-slide discussion and further team meetings to refine the final comprehensive TDPR and graduate extensions. Final submission deadline: see Canvas.

Other Disclosures

Modifications. The instructor may revise course activities or other details to improve the learning experience. Material changes will be communicated through Canvas.

Salutation. During the course, the instructor's preferred salutation is Prof. Aguilar.

Academic integrity. Students are expected to adhere to the Duke Community Standard and relevant university academic-integrity policies. Submitting another person's work as your own, misrepresenting authorship or data, undocumented use of sources, unauthorized collaboration on individual work, and misrepresenting AI-generated work may constitute academic dishonesty. Permitted AI use does not replace the obligation to understand, verify, and appropriately document the submitted work. Suspected violations will be addressed under university procedures.

The Duke Community Standard asks members of the community to commit to honesty, fairness, and accountability, including the pledges: "I will not lie, cheat, or steal in my academic endeavors"; "I will conduct myself honorably in all my endeavors"; and "I will act if the Standard is compromised."

Academic accommodations. Students seeking disability-related accommodations should register with Duke's Student Disability Access Office (SDAO). Accommodations can be implemented after the instructor receives the appropriate faculty accommodation letter. Contact: sdao@duke.edu.

Resources to support student success. Students facing difficulties with course preparation, academics, technology, finances, or personal well-being are encouraged to contact the instructor and appropriate Duke support services. Resources described in the university's materials include the Academic Resource Center, DukeReach, Counseling and Psychological Services (CAPS), and library and financial-aid programs. Current contact details and eligibility conditions are available through Duke's official channels.

Discussion guidelines. Listen respectfully, support claims with evidence, and give classmates room to contribute. Professional conduct affects participation.